

Computer Architecture and Organization

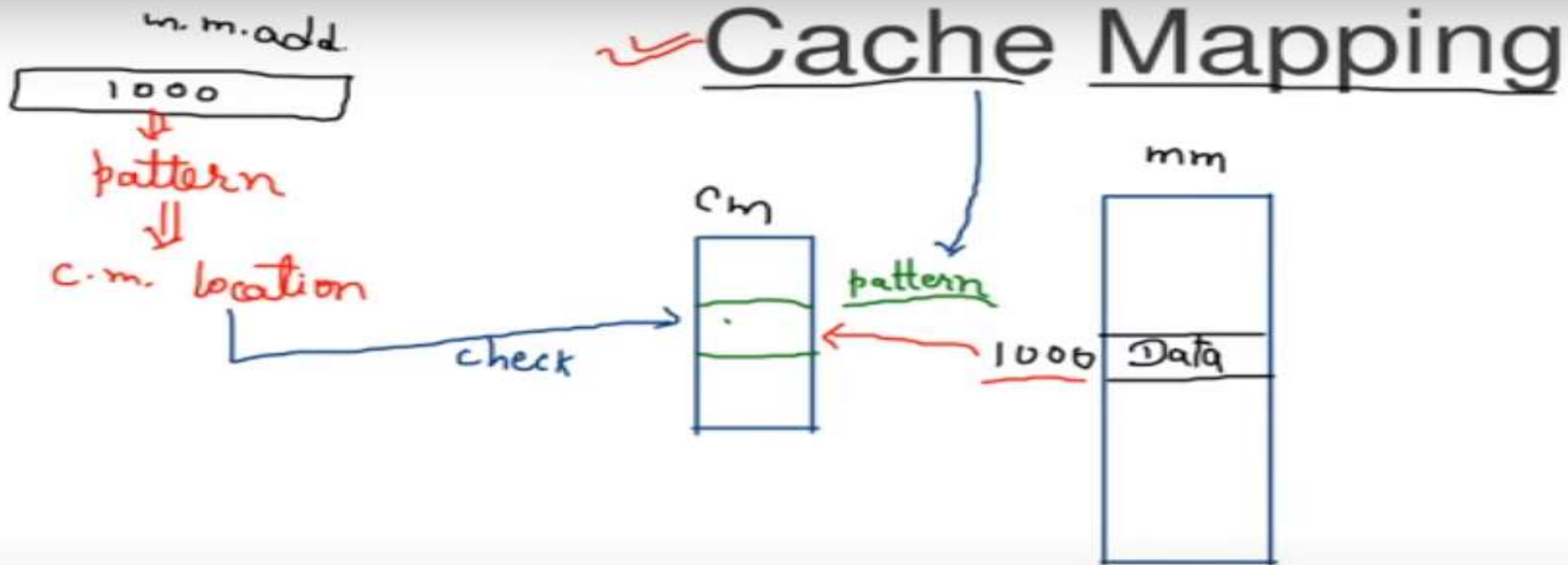
Lec 8

Memory-2

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Cache Mapping

- CPU always generates the address of main memory.



There is a pattern using which mm content from a specific location is copied in cm. at specific location.

Cache Memory

- The cache contains a copy of portions of main memory. When the processor attempts to read a word of memory, a check is made to determine if the word is in the cache. If so, the word is delivered to the processor. If not, a block of main memory, consisting of some fixed number of words, is read into the cache and then the word is delivered to the processor.
- Because of the phenomenon of locality of reference, when a block of data is fetched into the cache to satisfy a single memory reference, it is likely that there will be future references to that same memory location or to other words in the block.

Mapping Function

- Because there are fewer cache lines than main memory blocks, an algorithm is needed for mapping main memory blocks into cache lines
- Three techniques can be used:

Direct

- The simplest technique
- Maps each block of main memory into only one possible cache line

Associative

- Permits each main memory block to be loaded into any line of the cache
- The cache control logic interprets a memory address simply as a Tag and a Word field
- To determine whether a block is in the cache, the cache control logic must simultaneously examine every line's Tag for a match

Set Associative

- A compromise that exhibits the strengths of both the direct and associative approaches while reducing their disadvantages

Direct Mapping: Pros and Cons

- The direct mapping technique is simple and inexpensive to implement. Its main disadvantage is that there is a fixed cache location for any given block. Thus, if a program happens to reference words repeatedly from two different blocks that map into the same line, then the blocks will be continually swapped in the cache, and the hit ratio will be low (a phenomenon known as *thrashing*).

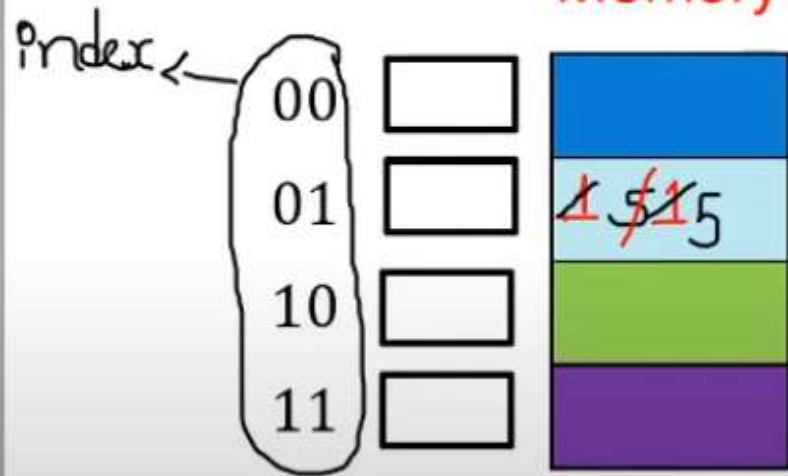
Problems of Direct Mapping

- Direct mapped cache replace a block whenever another memory reference needs that block.
- The block that is replaced is the victim block

Set Associative Mapping

Direct

Cache = 4 blocks
Memory



Main Memory = 8 blocks

0000	Blue
0001	Blue
0010	Light Blue
0011	Light Blue
0100	Green
0101	Green
0110	Purple
0111	Purple
1000	Blue
1001	Blue
1010	Light Blue
1011	Light Blue
1100	Green
1101	Green
1110	Purple
1111	Purple

CPU Req:-

mm blocks = 1, 5, 1, 5, 1, 5

miss

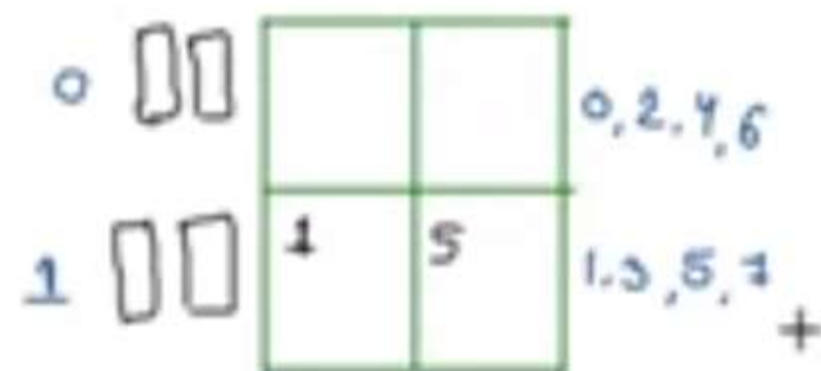
miss

miss

miss

Set Associative Mapping

2-way set associative:-



CPU Req:- 1, 5, 15, 15

miss miss hit



cm
set no.

$$= \frac{\text{mm block no.}}{\text{(no. of sets in cache)}}$$

$$\text{Tag directory size} = \text{No. of blocks in cache} * [\text{Tag-bits} + \text{extra-bits}]$$

$$\text{No. of sets in cache} = \frac{\text{No. of block in cache}}{\text{associativity}}$$

m.m. add.

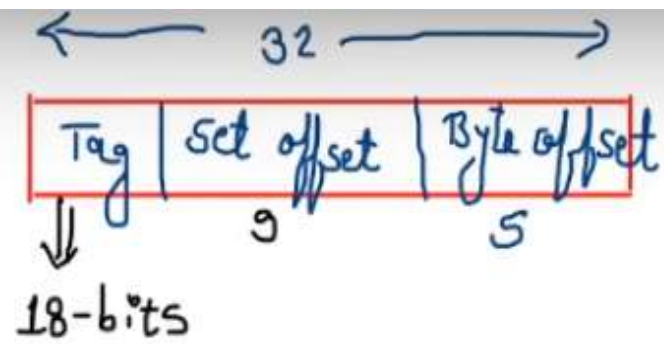


Ques:- mem add. = 32-bits
cache size = 64KB
Block size = 32B = 2^5B

4-way set associative cache

$$\text{No. of blocks in cache} = \frac{64KB}{32B} = 2K = 2^{11}$$

$$\text{No. of sets in cache} = \frac{2^{11}}{4} = 2^9$$



$$\begin{aligned}\text{Tag directory size} &= 2^{11} \times 18\text{-bits} \\ &= 36Kb\end{aligned}$$

Consider a 4-way set associative cache consisting of 128 lines with a line size of 64 words. The CPU generates a 20 – *bit* address of a word in main memory. The number of bits in the TAG, LINE and WORD fields are respectively:

A. 9, 6, 5

B. 7, 7, 6

C. 7, 5, 8

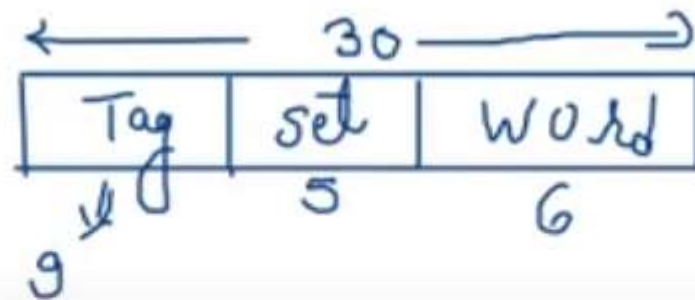
D. 9, 5, 6

Solution

$$\text{no. of blocks} = 128 = 2^7$$

$$\begin{aligned}\text{block size} &= 64 \text{ words} \\ &= 2^6 \Rightarrow \text{WORD} \\ &= 6\text{-bits}\end{aligned}$$

$$\text{no of sets} = \frac{2^7}{2^2} = 2^5 \Rightarrow \text{SET} = 5\text{-bits}$$



Consider a computer with a 4-ways set-associative mapped cache of the following characteristics: a total of 1 *MB* of main memory, a word size of 1 byte, a block size of 128 words and a cache size of 8 *KB*.

The number of bits in the TAG, SET and WORD fields, respectively are:

- A. 7, 6, 7 B. 8, 5, 7 C. 8, 6, 6 D. 9, 4, 7

Consider a computer with a 4-ways set-associative mapped cache of the following characteristics: a total of 1 MB of main memory, a word size of 1 byte, a block size of 128 words and a cache size of 8 KB.

The number of bits in the TAG, SET and WORD fields, respectively are:

A. 7, 6, 7

B. 8, 5, 7

C. 8, 6, 6

D. 9, 4, 7

$$\text{Main memory} = 1 \text{ MB} = 2^{20} \text{ B} \Rightarrow \text{main memory add.} = \underline{20\text{-bits}}$$

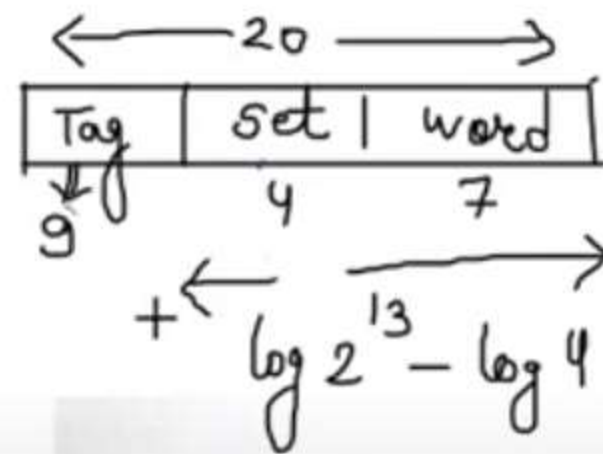
$$\text{Block size} = 128 \text{ words} = 2^7 \Rightarrow \text{word} = 7\text{-bits}$$

$$\text{Cache size} = 8 \text{ KB}$$

$$\text{No. of blocks} = \frac{8 \text{ KB}}{2^7} = \frac{2^{13}}{2^7} = 2^6$$

$$\text{No. of sets} = \frac{2^6}{4} = 2^2$$

SET = 4-bits



Consider a computer with a 4-ways set-associative mapped cache of the following characteristics: a total of 1 MB of main memory, a word size of 1 *byte*, a block size of 128 words and a cache size of 8 *KB*.

While accessing the memory location $0C795H$ by the CPU, the contents of the TAG field of the corresponding cache line is:

A. 000011000

B. 110001111

C. 00011000

D. 110010101

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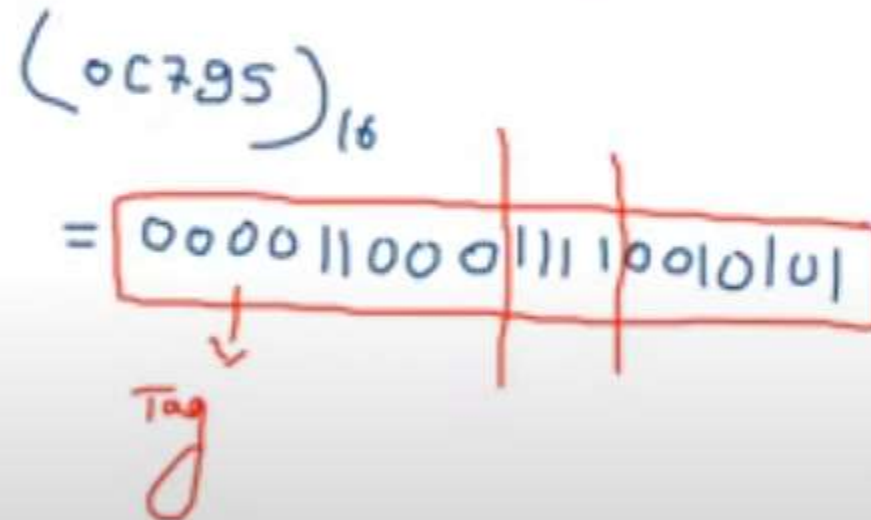
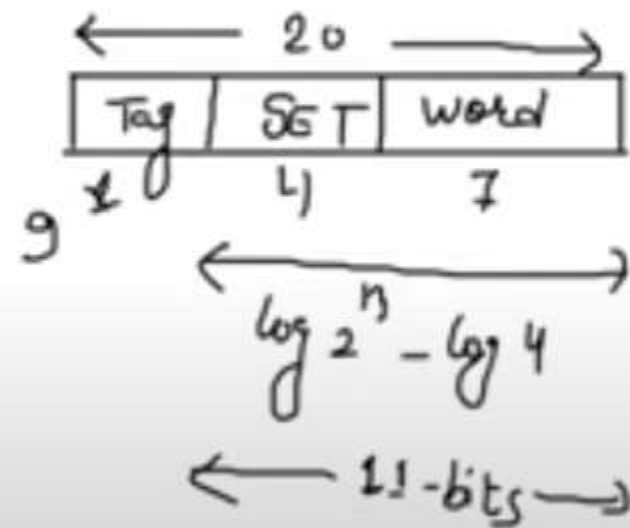
While accessing the memory location 0C795H by the CPU, the contents of the TAG field of the corresponding cache line is:

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Fully Associative Mapping